

Incremental Dockerfile Tutorial

This document introduces Dockerfile keywords step by step, with runnable examples. Each step adds new instructions while keeping the container functional.

Step: FROM & COPY

FROM defines the base image. COPY copies files from your system into the container.

```
FROM nginx:alpine
COPY . /usr/share/nginx/html
```

Step: LABEL

LABEL adds metadata to the image (e.g., maintainer, purpose).

```
FROM nginx:alpine
LABEL maintainer="you@example.com"
LABEL purpose="Teaching Docker keywords"
COPY . /usr/share/nginx/html
```

Step: WORKDIR

WORKDIR sets the working directory inside the container.

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
COPY . .
```

Step: ENV

ENV defines environment variables inside the container.

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
ENV NGINX_PORT=80
COPY . .
```

Step: RUN

RUN executes a command at build time (e.g., install software).

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
RUN apk add --no-cache curl
COPY . .
```

Step: EXPOSE

EXPOSE documents the port the container will listen on.

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
COPY . .
EXPOSE 80
```

Step: ENTRYPOINT & CMD

ENTRYPOINT defines the main command. CMD provides default arguments or command.

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
COPY . .
EXPOSE 80
ENTRYPOINT ["nginx", "-g"]
CMD ["daemon off;"]
```

Step: MULTI-STAGE

Multi-stage builds let you separate build and runtime images for efficiency.

```
FROM alpine AS builder
RUN echo "<h1>Hello from Multi-stage!</h1>" > /index.html
```

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
COPY --from=builder /index.html .
```

Step: VOLUME

VOLUME declares a mount point for persistent or shared data.

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
COPY . .
EXPOSE 80
VOLUME ["/usr/share/nginx/html"]
ENTRYPOINT ["nginx", "-g"]
CMD ["daemon off;"]
```

Step: ARG

ARG defines build-time variables (different from ENV).

```
FROM nginx:alpine
ARG build_version="1.0"
LABEL version=$build_version
WORKDIR /usr/share/nginx/html
COPY . .
EXPOSE 80
```

Step: USER

USER specifies which user runs the container process (security).

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
COPY . .
RUN adduser -D myuser
USER myuser
EXPOSE 80
```

Step: HEALTHCHECK

HEALTHCHECK defines a command to test container health.

```
FROM nginx:alpine
WORKDIR /usr/share/nginx/html
COPY . .
EXPOSE 80
HEALTHCHECK --interval=30s --timeout=3s CMD curl -f http://localhost/ || exit 1
ENTRYPOINT ["nginx", "-g"]
CMD ["daemon off;"]
```